UBER DATA BREACH (2016)

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CHAPTER 1

INTRODUCTION

The Uber data breach incidents have become a landmark case in cybersecurity, illustrating how a single vulnerability can set off a chain of recurring security failures if not managed with effective risk mitigation and incident response. This case study examines the major breaches Uber experienced, beginning with the initial data breach in 2014, which involved attackers gaining unauthorized access to sensitive information by exploiting exposed credentials on a public code repository. Despite early signs of vulnerabilities, Uber's security protocols remained insufficiently fortified, leading to a larger breach in 2016 exposing the personal information of 57 million Uber users and drivers worldwide.

This case study is conducted with reference to the principles covered in the CompTIA Security+ certification, focusing on core areas of security risk management and foundational cybersecurity principles. By using the framework of Security+ content, the analysis explores how security architecture, incident response, identity and access management, and data governance could have been applied to prevent or mitigate these breaches. The Security+ content provides a systematic lens through which Uber's security failures can be evaluated, highlighting critical security gaps and underscoring the importance of proactive security protocols and robust incident management.

The case study aims to analyse the breach, threats, vulnerabilities and outline how adherence to cybersecurity fundamentals could have transformed Uber's approach to risk management against the escalating cyber threats it faced.

1.1. COURSE => Udemy CompTIA Security+

CompTIA Security+ is a globally recognized certification that establishes a foundational understanding of core cybersecurity concepts and best practices. As one of the industry's most respected entry-level security certifications, it provides essential skills for identifying,

managing, and mitigating security risks in a wide range of IT environments. The course is designed to equip professionals with the knowledge and tools necessary to protect networks, data, and systems, making it highly valuable for roles such as security analysts, network administrators, and systems administrators.

The CompTIA Security+ course provides a comprehensive foundation in cybersecurity, covering essential areas such as risk management, identity and access control, and secure network architecture. Key principles include assessing and mitigating threats like malware, phishing, and Advanced Persistent Threats (APTs), along with the fundamentals of encryption, cryptography, and Public Key Infrastructure (PKI). It also emphasizes secure software development within the software development lifecycle (SDLC), secure cloud and virtualization practices, and Zero Trust architecture.

People learn to design, implement, and enforce security policies, manage incident response, and ensure business continuity and disaster recovery. The course rounds out with hands-on experience in monitoring, detection, threat hunting, and user awareness, promoting an in-depth understanding of data privacy and protection.

1.2. CASE STUDY => Uber Data Breach 2016

The Uber data breaches are an important case study in cybersecurity due to the sequence of incidents across multiple years, demonstrating how unresolved vulnerabilities can create a lasting impact. Starting with the initial breach in 2014, Uber faced a cascade of security challenges that were amplified in 2016 and again in 2019, each incident exposing the organization’s struggles with securing its data and systems. These breaches not only compromised sensitive information belonging to millions of users and drivers but also highlighted critical flaws in Uber’s security protocols and incident response procedures. Here is a breakdown of each incident, exploring the vulnerabilities exploited, how each breach occurred, and its lasting impacts.

1.2.1. **Uber Data Breach 2014:**

In May 2014, Uber experienced its first known data breach when attackers gained unauthorized access to Uber's internal system. The attackers were able to download a large volume of sensitive data, including the personal details of 50,000 Uber drivers.

The breach was possible due to an oversight in Uber's security practices. Uber developers had accidentally posted an access key to a public code repository on GitHub, which granted access to Uber's AWS storage. Once the attackers discovered the key, they used it to enter Uber's cloud environment and gain access to files containing sensitive driver data.

Vulnerabilities:

- Exposed credentials in a publicly accessible GitHub repository.
- Lack of secure coding practices
- Poor access control policies
- Failure to monitor and control access to its cloud storage AWS database

Impact:

The breach affected approximately 50,000 Uber drivers, exposing personal data such as names, driver's license numbers, and other identifying information. Although this breach initially went unreported for some time, it later caught the attention of authorities, resulting in fines and significant reputational damage. This incident was a warning sign of Uber's inadequate security practices.

1.2.2. **Uber Data Breach 2016:**

In October 2016, Uber suffered a far more extensive data breach, which compromised the personal information of approximately 57 million Uber users and drivers globally. This breach, however, remained hidden from the public for over a year, as Uber attempted to conceal it. Much like the 2014 incident, the breach was facilitated by weak access controls and compromised credentials. Attackers obtained credentials to Uber's cloud environment through a similar method, gaining unauthorized access to Uber's GitHub account and subsequently accessing an AWS S3 bucket containing sensitive data. Inside the bucket, they found files containing personal information on millions of users and drivers, which they proceeded to download.

Vulnerabilities:

- **Insufficient Access Controls**: Uber failed to properly secure its GitHub repositories and cloud environment, allowing attackers to locate access credentials with minimal effort.
- **Inadequate Encryption**: The data stored in the S3 bucket was not adequately encrypted, making it accessible to anyone who had access to the AWS account.

Impact:

The breach exposed a wealth of personal data, including names, email addresses, phone numbers of riders, and the driver's license numbers of Uber drivers. Instead of reporting the breach to the authorities as required, Uber paid the attackers $100,000 to delete the data and kept the incident under wraps. This concealment led to severe regulatory scrutiny, fines, and settlements costing the company millions. It also seriously damaged Uber's reputation and led to leadership changes.

CHAPTER 2

RISK ANALYSIS

Through risk analysis, companies can assess their security posture, identifying potential vulnerabilities, evaluating the likelihood and impact of threats, and implementing appropriate countermeasures. The security risks associated with the Uber data breach is as follows:

- **Technical Risk**

Insufficient access control policies and practices, along with exposed credentials in public repositories, created significant vulnerabilities. This allowed unauthorized individuals to access critical systems and sensitive data. Weak management of credentials and lack of multi-factor authentication further exacerbated the risk, leaving the organization's assets unprotected.

Mitigation Techniques:

- Implement multi-factor authentication (MFA) for all critical access points
- Enforce the principle of least privilege by updating access permissions
- Mandate encryption of sensitive data at all stages

- **Operational Risk**

Uber's delays in incident reporting and response allowed attackers more time within the system, which amplified the breach's impact. Concealing data breaches also violated compliance standards, attracting regulatory scrutiny and reputational damage. Additionally, limited security monitoring allowed unauthorized activity to go undetected for extended periods, increasing the duration and severity of the breach.

Mitigation Techniques:

- Develop and document a robust incident response plan
- Conduct regular incident response drills and post-incident evaluations
- Train employees to report incidents promptly and transparently

- **Third Party Vendor Risks**

Reliance on third-party vendors without rigorous security evaluations exposed Uber to additional risks through less secure channels. Inconsistent vendor security practices and lack of adequate monitoring heightened the potential for breaches originating from these external sources, thereby increasing Uber's vulnerability.

Mitigation Techniques:

- Enforce a vendor management program that includes security standards and regular audits
- Require third-party vendors to implement robust security protocols and access controls
- Develop a formalized risk assessment process before engaging with vendors

- **Employee Training and Social Engineering Risks**

Inadequate security awareness and training left Uber's employees vulnerable to social engineering attacks like phishing, which attackers exploited to gain unauthorized access to internal systems. Without sufficient training, employees were more susceptible to compromising organizational security inadvertently.

Mitigation Techniques:

- Implement ongoing security training programs focused on phishing, social engineering, and credential protection to build employee resilience against attacks.
- Run simulated phishing tests to assess and improve employee readiness and awareness
- Develop a security culture that encourages employees to report suspicious activity, promoting proactive security measures.

- **Compliance and Regulatory Risks**

Uber's inconsistent compliance with data protection regulations, along with weak governance practices, resulted in regulatory fines and reputational harm. Failure to adhere to compliance standards hindered the enforcement of security policies and exposed Uber to legal and financial repercussions.

Mitigation Techniques:

- Develop a compliance framework aligned with data protection regulations, such as GDPR
- Perform regular security and compliance audits to identify and mitigate gaps in governance and policy adherence.

- Appoint a compliance officer responsible for overseeing data protection and regulatory compliance, ensuring accountability within the organization.

2.1. **THREAT AND VULNERABILITY PAIR MATRIX**

The Threat-Vulnerability Pair Matrix shown outlines specific threats and the corresponding vulnerabilities that were exploited or exposed during the Uber data breach. It provides an analysis of the risks associated with each pair and potential mitigation measures. The Table 2.1. gives the Threat- Vulnerability pairs for the case study.

Table 2.1. The table shows the Threat -Vulnerability pairs of the Uber Breach Incident

Threat Actor	Threat Action	Vulnerability
Cybercriminals	Unauthorized access to sensitive data in third party cloud service (AWS)	Weak access
Cybercriminals	Unauthorized access to sensitive data	Exposed credentials
Negligent insiders	Uploaded sensitive credentials to GitHub repository	Insufficient employee training and monitoring
Insiders	Data manipulation or unauthorized access	Failure to follow security protocols
Third-Party Vendor Vulnerabilities	Exploitation of third-party vulnerabilities	Lack of continuous security monitoring

Third-Party Vendor Vulnerabilities	Exploitation of third-party vulnerabilities	Poor access controls
Phishers	Phishing attacks to gain access into internal system	Lack of employee awareness
Phishers	Phishing attacks to gain access into internal system	Insufficient employee training
Script kiddies	Using automated tools to exploit known vulnerabilities	Unpatched systems
Uber management	Covered up data breach and delayed incident response	Weak compliance and governance
Uber management	Covered up data breach and delayed incident response	Poor incident response plan

2.2. IMPACT SCORE

The impact score for each threat and vulnerability pair is evaluated by the average of the CIA triad scores and ranges from 1 to 5:

- 1 = Low impact
- 2 = Moderate impact
- 3 = Significant impact
- 4 = High impact
- 5 = Critical impact

This scoring system is based on the potential damage that the threat may inflict, taking into account various factors such as data loss, operational disruptions, reputational harm, and regulatory repercussions. The Table 2.2. gives the impact score for each threat.

Table 2.2. The table gives the impact score for each threat

Threat	Vulnerability	Confidentiality Score	Integrity Score	Availability Score	Impact Score
Cybercriminals Unauthorized access to sensitive data in third-party cloud service	Weak access controls	5	4	3	4.00
Cybercriminals Unauthorized access to sensitive data in third-party cloud service	Exposed credentials	5	3	2	3.33
Negligent insiders Uploaded sensitive credentials to GitHub repository	Insufficient employee training and monitoring	4	5	2	3.6
Negligent insiders Uploaded sensitive credentials to GitHub repository	Failure to follow security protocols	4	5	3	4.0
Third-Party Vendor Vulnerabilities Exploitation of third-party vulnerabilities	Lack of continuous security monitoring	5	3	4	4.00

Third-Party Vendor Vulnerabilities Exploitation of third-party vulnerabilities	Poor access controls	5	4	3	4.00
Phishers Phishing attacks to gain access to internal systems	Lack of employee awareness	4	2	3	3.00
Uber management Covered up data breach and delayed incident response	Weak compliance and governance and Poor incident response plan	5	5	4	4.67

2.3. RISK SCORE

Each identified risk is evaluated with three primary metrics:

1. **Impact score** (1 to 5): Represents the severity of the threat
2. **Likelihood score** (1 to 5): Indicates the probability of its occurrence
3. **Risk score** = Impact Score × Likelihood.

The Table 2.3. gives the risk scores obtained from the impact and likelihood score.

Table 2.3. The table shows the estimated Risk score for each threat

Threat	Vulnerability	Impact Score	Likelihood Score	Risk Score (Impact*Likelihood)
Cybercriminals Unauthorized access to sensitive data in third-party cloud service	Weak access controls	4.00	4	16

Cybercriminals Unauthorized access to sensitive data in third-party cloud service	Exposed credentials	3.33	4	13.32
Negligent insiders Uploaded sensitive credentials to GitHub repository	Insufficient employee training and monitoring	3.6	3	10.80
Negligent insiders Uploaded sensitive credentials to GitHub repository	Failure to follow security protocols	4.0	3	12.00
Third-Party Vendor Vulnerabilities Exploitation of third-party vulnerabilities	Lack of continuous security monitoring	4.00	4	16.00
Third-Party Vendor Vulnerabilities Exploitation of third-party vulnerabilities	Poor access controls	4.00	4	16.00
Phishers Phishing attacks to gain access to internal systems	Lack of employee awareness	3.00	4	12.00
Uber management Covered up data breach and delayed incident response	Weak compliance and governance and Poor incident response plan	4.67	3	14.01

CHAPTER 3

SECURITY RISK MANAGEMENT STRATEGIES

Security risk management strategies are essential for organizations to identify, assess, and mitigate potential threats that can compromise their assets, operations, and reputation.

- **Third-Party Vendor Risk Management**

Third-party vendor risk management involves assessing and managing the risks associated with external vendors who have access to an organization's sensitive data and systems. Poor vendor management can lead to vulnerabilities that may be exploited, as seen in the Uber data breach.

Mitigation Techniques:

- **Vendor Assessment:** Conduct thorough due diligence on potential vendors, including security practices, compliance with industry standards, and previous breach histories before engagement.
- **Contractual Obligations:** Ensure contracts with vendors include clear security requirements, liability clauses, and breach notification procedures.
- **Regular Audits:** Schedule periodic security audits and assessments of vendors to ensure ongoing compliance with contractual obligations and security standards.
- **Access Controls:** Limit vendor access to sensitive systems and data strictly to what is necessary for their services.

- **Identity and Authorization Management**

Identity and authorization management encompasses the processes and technologies used to manage user identities and control access to resources based on defined permissions. Effective IAM helps prevent unauthorized access and ensures that users can only perform actions necessary for their roles.

Mitigation Techniques:

- **Strong Authentication Mechanisms:** Implement multi-factor authentication (MFA) and strong password policies to enhance user authentication and reduce the risk of credential theft.
- **Role-Based Access Control (RBAC):** Utilize RBAC to assign permissions based on user roles, ensuring users have the minimum necessary access to perform their job functions.
- **Regular Access Reviews:** Conduct regular reviews of user access permissions to ensure they remain appropriate as roles change or when employees leave the organization.
- **Identity Lifecycle Management:** Establish processes for the onboarding, role changes, and offboarding of users to manage identities effectively and maintain security.

- **Alerting and Monitoring**

Alerting and monitoring involve continuous oversight of systems and networks to detect and respond to security incidents. Effective monitoring helps organizations identify potential threats and mitigate risks before they escalate into significant breaches.

Mitigation Techniques:

- **Log Aggregation:** Centralize log data from various sources for comprehensive analysis, enabling better visibility into system events and potential security incidents.
- **Real-Time Alerts:** Set up automated alerts for suspicious activities, such as unusual login attempts or access to sensitive data outside of normal hours.
- **Regular Vulnerability Scanning:** Implement regular scans of systems and applications to identify vulnerabilities and misconfigurations that could be exploited by attackers.
- **Incident Response Plans:** Develop and maintain an incident response plan to guide the organization in quickly responding to detected incidents and mitigating damage.

3.1.THIRD-PARTY VENDOR RISKS MANAGEMENT

In the Uber data breach case study, one of the primary third-party vendor risks involved was the exposure of AWS credentials, which enabled unauthorized access to sensitive data stored on Uber's cloud platform.

- **Vendor Assessments**
  - **Pre-Partnership Assessment**: Evaluate a vendor's security posture before partnership, assess AWS's credential management practices, access controls, and cloud security protocols. Further, identify potential vulnerabilities in vendor access management and verify implementation of multi-factor authentication (MFA) and regular audits of user access.
  - **Automated Credential Scanning:** Implement tools to automatically scan for exposed credentials in public repositories (e.g., GitHub). Try to reduce the window of opportunity for attackers, as experienced in the Uber breach with AWS keys by using real-time alerts to notify security teams of sensitive credentials found, enabling rapid response.
  - **Audit Rights:** Establish audit rights in vendor contracts for regular evaluations of AWS's security practices. Allow Uber to review AWS's internal security processes and assess access controls and ensure incident response protocols are defined and effective.

- **Vendor Selection and Monitoring**
  - **Continuous Monitoring:** Use 24/7 security tools, like Security Information and Event Management (SIEM) systems, track and analyze activities within the AWS environment for unauthorized access attempts.
  - **Anomaly Detection Systems:** Implement systems to flag unusual behaviors or access patterns within AWS resources. Use advanced analytics to detect deviations from established norms (e.g., unexpected access or data transfer activities). Al**so,** ensure quick intervention when suspicious activities are identified to prevent exploitation of compromised credentials.
  - **Regular Security Reviews:** Conduct ongoing assessments of third-party vendors' security practices. Evaluate AWS's security measures, compliance with standards,

and results of audits or vulnerability assessments. Identify potential risks and enforce corrective actions proactively.

3.2.IDENTITY AND AUTHORIZATION MANAGEMENT

Implementing a robust IAM strategy is crucial for Uber to secure its sensitive data and resources. Risks can be reduced significantly through this approach:

- **Establish Robust IAM Framework:** Ensure that the right individuals have the right access to the right resources for the right reasons. Implement a comprehensive IAM framework that incorporates essential components like password management, network access control, and digital identity management. Regularly review and update IAM policies to reflect changes in the threat landscape and organizational structure.

- **Strengthen Password Management:** Prevent unauthorized access through effective password policies. Enforce strong password policies that require complexity, length, and regular updates. Implement password managers to assist users in generating and storing secure passwords.

- **Implement Multi-Factor Authentication (MFA):** Enhance user authentication and reduce the risk of credential theft. Utilize a combination of authentication factors, including:
  - **Something you know:** Strong passwords or PINs.
  - **Something you have:** Security tokens, mobile devices, or hard tokens.
  - **Something you are:** Biometrics (e.g., fingerprints, facial recognition).
  - **Something you do:** Behavioral biometrics.
  - **Somewhere you are:** Location-based authentication.

- **Robust IAM Processes (IAAA):** Ensure comprehensive identity management through well-defined processes.
  - **Identification:** Ensure all users claim unique identifiers

- **Authentication:** Implement strong verification mechanisms to confirm user or device
- **Authorization:** Clearly define user permissions post-authentication to limit access
- **Accounting:** Maintain detailed logs of user activities to track access patterns

- **Continuous Monitoring and User Behavior Analytics:** Detect and respond to unusual user behaviors that may indicate security threats. Implement continuous monitoring systems to analyze user activities and access patterns. Establish behavioral baselines to identify deviations that could indicate compromised accounts or insider threats. Utilize Security Information and Event Management (SIEM) tools to aggregate logs and detect anomalies in real time.

- **Access Control Models:** Regulate user access based on defined criteria to mitigate risks.
  - **Role-Based Access Control (RBAC):** Assign access rights based on user roles to minimize unnecessary access.
  - **Attribute-Based Access Control (ABAC):** Implement dynamic access controls based on attributes such as location, time, and device to ensure context-aware access.
  - Regularly review and adjust access permissions to align with changes in roles and responsibilities.

CHAPTER 5

AUDIT TABLE

Company Name: Amazon Inc.

Location: Amazon Headquarters, 410 Terry Ave N, Seattle, WA 98109, United States

Time Frame: December 2021

Table 4.1: This table provides an overview of audits conducted at Amazon Inc.

Checklist Item	**Exists?**	**Comments / Observations**
1. Software Supply Chain Risk Management		
a. Are evaluations conducted on third-party software dependencies?	No	Risk evaluations of third-party vendors are not conducted consistently.
b. Is there active monitoring of high-risk software (e.g., SolarWinds)?	Yes	Routine monitoring is established, though lacks extensive logging analysis.
c. Are contractual obligations in place for vendor incident reporting?	No	Incident disclosure requirements for vendors are not part of contractual agreements.
2. Patch Process Management		
a. Is there a defined, timely policy for patch management?	Yes	A policy exists, though implementation occasionally faces delays.
b. Are updates for high-risk applications reviewed prior to deployment?	No	There is no standardized review process before deploying high-risk application updates.
3. Access Control & Privilege Management		

a. Are privileged accounts consistently monitored?	Yes	Monitoring is in place, though lacks sufficient automation for alerts.
b. Are least privilege principles implemented for high-risk application access?	No	Numerous users possess elevated access rights to critical applications unnecessarily.
4. Network Segmentation and Monitoring		
a. Is there network isolation for critical applications?	No	High-risk software and critical systems share network segments, increasing security exposure.
b. Is traffic monitored for unusual patterns on critical segments?	Yes	Monitoring is performed, yet anomaly detection rules require enhancement.
5. Security Information and Event Management (SIEM)		
a. Does SIEM generate alerts, and are they actively monitored?	Yes	Alerts are actively monitored, though frequent false positives cause monitoring fatigue.
b. Are detection rules configured in the SIEM for critical application activities?	No	There are no specific detection rules to flag unusual activities related to high-risk software.
6. Incident Response and Forensics		
a. Is an incident response plan (IRP) established?	Yes	Documented IRP exists but is not consistently updated with the latest threat intelligence.
b. Are regular tests conducted on the IRP?	No	The last test was over a year ago, which could impact readiness in crisis scenarios.
7. Threat Intelligence and Alerts		

a. Is threat intelligence integrated into security operations workflows?	No	Threat feeds are updated manually, which slows response efforts.
b. Did the organization receive and act upon alerts (e.g., CISA alerts) for high-risk applications?	Yes	Alerts were acknowledged, though response times were noted to be slow.
8. Endpoint Detection & Response (EDR)		
a. Are high-risk servers included in EDR coverage?	Yes	Monitoring is active; however, alert sensitivity requires optimization.
b. Are suspicious endpoint events investigated in a timely manner?	No	Resource constraints lead to delays in investigating suspicious activity.
9. Employee Awareness & Training		
a. Are employees educated on recognizing supply chain risks?	No	Training lacks specific focus on supply chain security vulnerabilities.
b. Are privileged users trained on secure use of high-risk tools?	Yes	Training is provided, though certification requirements for privileged users are not enforced.
10. Vulnerability Assessments & Penetration Testing		
a. Are regular vulnerability assessments performed?	Yes	Vulnerability scans are routinely conducted, but some critical assets are omitted.
b. Are both internal and external penetration tests conducted regularly?	No	The last penetration test occurred over 18 months ago, potentially limiting threat visibility.

CHAPTER 5

SECURITY RISK MANAGEMENT RECOMMENDATION

Security Risk Management for the Uber data breach highlights critical recommendations based on CompTIA Security+ principles:

- **Strengthen Identity and Access Management (IAM):** To Implement robust IAM protocols, including multi-factor authentication (MFA), strong password policies, and least privilege access controls. Stronger IAM practices would reduce the likelihood of unauthorized access to sensitive data.

- **Develop and Enforce a Data Governance Policy**: To establish a comprehensive data governance policy that defines data classification, encryption standards, and secure storage requirements for sensitive information. Properly classifying and protecting data could prevent unauthorized access and misuse.

- **Incident Response and Breach Disclosure Protocol**: To create a transparent incident response plan that includes prompt breach disclosure to affected parties and regulators. An established protocol for rapid incident response, including responsible disclosure, can mitigate reputational and legal risks.

- **Proactive Employee Training and Cybersecurity Awareness:** To establish ongoing cybersecurity training for employees, focusing on secure coding practices, data handling, and recognizing potential social engineering attacks. Training would minimize the risk of errors that lead to vulnerabilities, such as accidentally exposing credentials.

CHAPTER 6

CONCLUSION

The Uber data breach case is a stark reminder of how lapses in cybersecurity can have profound and far-reaching consequences. Despite early warning signs, Uber's insufficient security measures and questionable response strategies allowed multiple breaches to occur, each compounding the damage to its reputation, finances, and the privacy of millions of users and employees. This case underscores the critical role of foundational cybersecurity practices—such as strict identity and access management, data governance, regular vulnerability assessments, and transparent incident response protocols—in safeguarding an organization's digital assets.

Implementing the recommended risk management strategies would not only have protected Uber from these high-profile breaches but also set a standard for responsible data stewardship. These strategies, in the CompTIA Security+ framework, offer a roadmap for strengthening cybersecurity resilience across all levels of an organization. In today's digital era, cybersecurity is not just an operational necessity but a fundamental aspect of maintaining user trust and upholding corporate responsibility. The lessons from Uber's experience should serve as a call to action for all organizations to prioritize and continuously improve their security practices, ensuring that they are equipped to address the challenges of an increasingly complex threat landscape.

The Uber data breaches highlight how cybersecurity failures can quickly escalate from technical issues to significant business crises, affecting not only the company's bottom line but also its public image and customer trust. By neglecting essential security controls and opting to conceal incidents rather than address them transparently, Uber not only breached legal and ethical standards but also missed the opportunity to turn a crisis into a learning and growth experience. A more robust approach to security, encompassing routine assessments, employee training, and a culture that prioritizes transparency and accountability, could have prevented or minimized these incidents.

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